

The JANUS Decision: Joint Adjudication via Negative-and-positive control Unified Signals for Observational Causal Inference

Raphael Derecki^{1,2}, Bogna Liziniewicz¹, James Goulding¹, and Brian O’Shea^{1,2}

¹ N/LAB, Nottingham University Business School, University of Nottingham,
Nottingham, UK

² School of Psychology, University of Nottingham, Nottingham, UK
`Raphael.Derecki@nottingham.ac.uk`

Abstract. Observational causal inference increasingly relies on auxiliary control outcomes—negative controls to detect residual confounding and positive controls to verify recovery of known effects. These literatures have developed independently, and neither specifies how the *joint* signature of both control types should adjudicate a target estimate. We unify this logic into a single pre-specified decision rule: the *JANUS Decision* (Joint Adjudication via Negative-and-positive control Unified Signals), which maps the combined pass/fail signature to one of four credibility verdicts—*release*, *recalibrate*, *redesign*, or *withhold*—each prescribing a distinct action, with severity tiers anchored in existing sensitivity-analysis and calibration machinery. The framework is estimator-agnostic and intended as a teachable diagnostic for researchers entering observational causal inference from applied backgrounds. We illustrate it on a sugar-sweetened-beverage and type-2-diabetes analysis using double machine learning, where the joint signature lands in the recalibrate cell and a literature-anchored bias correction reduces the headline estimate by 7.7% while leaving it significant; three constructed variants populate the remaining cells. Because the verdict is fixed before the target estimate is seen, JANUS replaces post-hoc, selectively reported robustness checks with a transparent rule researchers can commit to in advance.

Keywords: causal inference · anomaly detection · negative and positive controls · credibility assessment · pre-specified decision rule · root cause analysis

1 Introduction

Observational causal inference routinely uses estimators substantially more flexible than the linear regressions of two decades ago. Double machine learning, causal forests, and targeted learning enable estimation in high-dimensional and heterogeneous-effect settings [2], and are deployed across numerous scientific fields. Validation infrastructure for the estimates these methods produce has not kept pace. One contemporary response is the use of negative controls —

outcomes or exposures with no plausible causal pathway from the target exposure but with shared confounding structure [11, 16]. These have become a staple of observational pharmacoepidemiology, spawning methodological extensions across causality theories [8, 16]. Positive controls — outcomes with established causal effects of the exposure, against which an estimator’s recovery of known signal can be checked — are applied less consistently; despite their canonical role in laboratory science, they have not received parallel formalisation in observational practice.

Even where deployed, these signals are used inconsistently. Placebo tests are a classic example of negative controls, and a pre-registered audit of them in leading economics journals found far fewer significant results than chance would predict, implying that placebo tests which should halt or qualify a causal claim are selectively withheld from publication [5]. Joint use of negative and positive controls is rarer still, and where it occurs it is most often population-level calibration — adjusting confidence intervals against a panel of known nulls and synthesised non-nulls — rather than a pre-specified rule returning a structured verdict over an individual estimate. This paper addresses that gap.

Several adjacent traditions operationalise the joint-control logic in part. The OHDSI consortium has developed the most mature framework for empirical calibration of observational studies, using panels of negative controls with synthesised positive controls to calibrate confidence intervals in pharmacoepidemiology [15]; its recent study-validity extension, however, relies primarily on negative controls, with positive controls retaining a calibration role [4]. Jeunen et al. (2024) [9] classify past A/B experiments into A/A and known-outcome classes, using the joint signature to validate metrics rather than individual treatment effects. Hunter et al. (2025) [8] give a unified potential-outcomes formalisation of negative and positive controls, targeting experimental rather than observational settings. Outside causal inference, safety-critical engineering deploys paired-check logic with hard withhold thresholds in redundant sensor architectures [10], where any joint-signature failure forces an immediate safe-state transition. Each operationalises a fragment of the logic; none integrates it into a single pre-specified decision rule returning a structured verdict over an individual observational estimate.

To achieve this, we propose the JANUS decision (Joint Adjudication via Negative-and-positive control Unified Signals): an adjudication function that maps whether pre-specified negative and positive controls pass or fail one of four credibility verdicts — *release*, *recalibrate*, *redesign*, or *withhold* (see Section 3.2 for more details). Applicable to any causal estimation approach, including double machine learning, causal forests, and conventional regression alike, JANUS is estimator-agnostic. Researchers pre-specify decision thresholds before examining the target estimate, apply the framework, and receive one of four recommendations: the estimate is released as decision-grade, recalibrated against a characterised bias mechanism, returned for estimator redesign, or withheld entirely. The contribution is one of synthesis, rather than methodological innovation: existing traditions each operationalise a fragment of the joint-control

logic, but none provides a unified pre-specified decision rule for individual observational causal estimates. The framework is intended to be a transparent and teachable approach, particularly for researchers entering observational causal inference from applied data science, machine learning, and behavioural-science backgrounds where OHDSI-style empirical calibration is not part of standard training.

We illustrate the JANUS framework using a causal inference analysis of sugar-sweetened-beverages and type-2-diabetes (in Section 4), then present constructed examples demonstrating each verdict (Section 5), and finally demonstrate how the framework can be applied across four application domains with different decision requirements (Section 6). The framework’s logic is fixed, while the thresholds and controls chosen are domain-dependent.

2 Background and Notation

2.1 Negative and Positive Control Outcomes in Potential Outcomes

Let A denote a target exposure, such as smoking or Sugar-sweetened beverage consumption (binary or continuous), Y the target outcome of substantive interest, such as lung cancer or diabetes. The quantity $\theta := \mathbb{E}[Y(1) - Y(0)]$ represents the average causal effect — that is, the expected difference in the outcome if everyone were exposed versus if no one were exposed — which the analyst aims to estimate from observational data. Let $\hat{\theta}$ denote the estimated average causal effect obtained from an observational study using a chosen estimator (e.g., causal forest), assumed consistent for θ under the analyst’s stated identification assumptions.

The framework below is developed for negative control *outcomes* (NCOs) rather than negative control *exposures*. NCOs dominate applied practice because of availability; most observational data sources record many outcomes during the same exposure making it feasible to identify outcomes expected on subject-matter grounds to be unaffected by A [16]. The adjudication logic applies equally to negative control exposures, with the identification conditions below reformulated for the exposure case.

A *negative control outcome* Y_{NC} is an auxiliary outcome the analyst posits, on subject-matter grounds, to satisfy two conditions: (i) the average causal effect of A on Y_{NC} should be zero, and (ii) any unmeasured confounder U affecting the $A \rightarrow Y$ relationship also affects the $A \rightarrow Y_{\text{NC}}$ relationship [11, 16]. The estimand $\beta_{\text{NC}} := \mathbb{E}[Y_{\text{NC}}(1) - Y_{\text{NC}}(0)]$ is therefore zero under correct identification, and any $\hat{\beta}_{\text{NC}}$ will be detectably (significantly) distinct from zero beyond sampling variation, produced by the same estimator on the same data, is interpretable as a confounding signature for the target estimate [8]. The diagnostic depends on (i) and (ii) holding: a misidentified Y_{NC} that is in fact causally affected by A will produce a non-null estimate even when the target estimator is unbiased, leading the analyst to incorrectly diagnose confounding. Subject-matter justification of the NCO choice is therefore a load-bearing step, as is its confounding sensitivity:

a valid NCO whose association with U is near zero satisfies (i) and (ii) yet has little power to surface confounding even when present.

A *positive control outcome* Y_{PC} is an auxiliary outcome the analyst posits to have a literature-established average causal effect θ_{PC}^* , available from prior randomised trials, meta-analyses, or experimental evidence external to the present analysis. The estimand $\beta_{PC} := \mathbb{E}[Y_{PC}(1) - Y_{PC}(0)]$ is expected to recover θ_{PC}^* within sampling variation, and any substantial deviation of $\hat{\beta}_{PC}$ from θ_{PC}^* is interpretable as evidence of estimator misspecification or of identification failure that prevents recovery of known signal [8].

The asymmetry between NCO and PC is encoded in their estimands: the NCO satisfies $\beta_{NC} = 0$ under correct identification, the PC satisfies $\beta_{PC} = \theta_{PC}^*$. Both are validation targets rather than experimental control conditions in the design sense; the JANUS decision is built on their joint pass/fail signature.

Two PC traditions coexist in the literature: *literature-anchored* PCs, where the expected effect comes from prior empirical evidence (e.g., meta-analytic anchors), and *synthesised* PCs, where known effects are injected into the data [15]. JANUS as instantiated here uses literature-anchored PCs; the framework applies equally to synthesised PCs, with the OHDSI calibration machinery substituted at the relevant severity tiers.

2.2 The JANUS Decision as a Formal Object

The JANUS decision is an adjudication function

$$J(\hat{\theta}, \hat{\beta}_{NC}, \hat{\beta}_{PC}; \tau_{NC}, \tau_{PC}, \theta_{PC}^*) \mapsto \{\text{release, recalibrate, redesign, withhold}\}, \quad (1)$$

where $\tau_{NC}, \tau_{PC} > 0$ are tolerance values and θ_{PC}^* is the literature-established reference effect (or, in the synthesised-PC case, the injected effect size). The arguments $\hat{\theta}, \hat{\beta}_{NC}, \hat{\beta}_{PC}$ may be obtained from any consistent causal-effect estimator applied to the same dataset.

The pre-specification requirement is central: $\tau_{NC}, \tau_{PC}, \theta_{PC}^*$ must be committed before $\hat{\theta}$ is unblinded; this does not require pre-registration or experimental studies. Without pre-specification, the rule degrades to a post-hoc robustness check interpretable in light of the target estimate — the empirical audit literature has shown such checks to be selectively reported in observational practice [5], and the OHDSI tradition has identified this as a barrier to trustworthy real-world evidence [4]. Section 3 develops the pass/fail definitions and the four-cell decision rule in full.

3 The JANUS Decision

3.1 Pass/Fail Conditions and the Adjudication Function

The JANUS decision adjudicates the joint pass/fail signature of two pre-specified checks on the auxiliary estimates $\hat{\beta}_{NC}$ and $\hat{\beta}_{PC}$ detailed in Section 2, and defined

as:

$$\text{NC pass} \iff |\hat{\beta}_{\text{NC}}| \leq \tau_{\text{NC}} \quad (2)$$

$$\text{PC pass} \iff |\hat{\beta}_{\text{PC}} - \theta_{\text{PC}}^*| \leq \tau_{\text{PC}} \quad (3)$$

The two tolerances encode different criteria. The negative-control tolerance τ_{NC} is a *negligibility bound*: the largest deviation of $\hat{\beta}_{\text{NC}}$ from its null value of zero that the analyst is willing to treat as consistent with negligible residual confounding. The positive-control tolerance τ_{PC} is an *agreement bound*: the largest discrepancy between $\hat{\beta}_{\text{PC}}$ and the literature-established effect θ_{PC}^* that the analyst is willing to treat as successful recovery of known signal. Both are fixed in advance on subject-matter grounds, τ_{NC} from the magnitude of confounding judged tolerable and τ_{PC} from the precision of the external estimate of θ_{PC}^* .

The conditions are stated as deviations against pre-specified numerical thresholds rather than confidence-interval tests or Bayesian posterior probabilities. Mapping continuous estimates to binary pass/fail discards information, but makes the resulting decision rule explicitly pre-specifiable. This matches the pre-specified gating of the OHDSI study-validity-diagnostics tradition [4], where thresholds are decided ahead of time, and the rule is calibrated against them rather than a data-dependent inferential procedure. More elaborate formulations are noted in Section 3.3; our contribution is the adjudication structure rather than a particular pass/fail formalism.

For multiple controls, JANUS operates on pre-specified panel-level summaries. The analyst therefore specifies how each panel is aggregated and its decision threshold before examining the target estimate. Aggregation should not conceal substantial within-panel disagreement: discordant controls may instead indicate that assumed shared confounding structure or external anchors are inappropriate, requiring diagnostic investigation before adjudication. The four-cell rule is then applied to the resulting panel-level NC and PC states.

3.2 The Four Verdicts

Release (NC pass, PC pass). The estimator produces a null effect on the negative-control outcome and recovers the literature-established effect on the positive-control outcome, each within its pre-specified tolerance. Both auxiliary signals are consistent with correct identification. *Action:* release $\hat{\theta}$ with normal scientific caveats. Passing the joint check does not guarantee correct identification, only that the auxiliary signals provide no evidence against it, conditional on the NC and PC choices being substantively appropriate. This pattern has precedent across large-scale online experimentation platforms [9, 23].

Recalibrate (NC fail, PC pass). The estimator recovers known signal but produces a non-null effect on the negative-control outcome, indicating characterisable confounding: a bias source affecting both the target and the NC, but not severe enough to prevent recovery of the established PC effect. *Action:* apply a bias correction informed by the NC effect, characterise the bias mechanism via

root-cause analysis, and release the recalibrated estimate with explicit residual-bias reporting. The simplest correction subtracts the NC effect from the target, $\hat{\theta}_{\text{adj}} = \hat{\theta} - \hat{\beta}_{\text{NC}}$, under the strong assumption that the bias acts identically on the target and NC variables [11]. More sophisticated calibrations relax this by fitting an empirical null distribution to a panel of NCs and shifting the target estimate by the fitted location parameter [15]; The severity tiers in Section 3.3 map the calibration method to the magnitude of the detected NC failure. In moderate-tolerance regimes the recalibrate verdict remains available; in safety-critical regimes it collapses to withhold, where any failure of the joint signature triggers an immediate safe-state transition rather than correction. Such regimes arise wherever acting on a biased estimate is itself costly or irreversible—for example, a pharmacovigilance signal feeding a regulatory decision—mirroring fault response in engineered safety systems, where a detected fault forces reversion to a known safe state (as in a reactor SCRAM) [10].

Redesign (NC pass, PC fail). The estimator produces a null effect on the negative-control outcome but fails to recover the literature-established positive-control effect: no confounding signature in the NC, so the failure is on the estimator side. *Action:* return to the design phase. Possible causes include estimator misspecification, inadequate covariate adjustment, or design-assumption violations such as no treatment interference [17]. The credibility verdict is low: a procedure that cannot recover known signal cannot be assumed to have recovered the unknown target effect either, even when the negative-control signal does not indicate confounding.

Withhold (NC fail, PC fail). Both auxiliary signals fail: confounding is present and the estimator cannot recover known signal. The joint signature cannot disambiguate the bias from the estimator failure, and provides no support for any version of $\hat{\theta}$. *Action:* withhold $\hat{\theta}$; the estimate must not propagate to downstream decisions. Possible next steps include: *redesign from first principles*, *re-scoping to a different causal question for which adequate NC and PC are available*, and *additional diagnostic work to disambiguate the failure modes*—for instance, sensitivity analysis quantifying the unmeasured confounding required to produce the observed negative-control deviation [20], which separates genuine confounding from a misidentified control. The estimate should be withheld. All regimes reach this verdict for the same reason: a failed positive control means the estimator cannot recover known signal, so there is no sound estimator left to recalibrate, while a failed negative control signals residual confounding—together leaving nothing for a higher tolerance to correct toward.

Unlike a binary release/withhold rule, the four-verdict framework makes the underlying diagnostic outcome explicit: each verdict communicates both the diagnostic conclusion and the appropriate action, rather than leaving the analyst to interpret the negative- and positive-control results separately.

Table 1. The JANUS adjudication. Agreeing cells (RELEASE, WITHHOLD) give unambiguous verdicts; disagreeing cells (RECALIBRATE, REDESIGN) carry the diagnostic load — the kind of failure sets the action.

	PC pass	PC fail
NC pass	Release	Redesign
NC fail	Recalibrate	Withhold

3.3 Setting the Thresholds

The JANUS rule requires two thresholds, τ_{NC} and τ_{PC} , partitioning the joint NC/PC signature into the four verdict cells. These are the one component the framework does not determine: neither has a derived optimal value. Each threshold trades off two opposing errors. A *false flag* occurs when a control fails despite sound identification—for instance, $|\hat{\beta}_{\text{NC}}|$ exceeding τ_{NC} through sampling variation rather than genuine confounding—and leads the analyst to withhold or recalibrate a sound estimate. A *false pass* occurs when both controls pass despite compromised identification—for instance, residual confounding that the negative control lacks the power to surface (Section 2)—and leads to release of a biased estimate. Lowering a threshold reduces false passes at the cost of more false flags, and raising it does the reverse; the optimal balance depends on the cost structure of the downstream decision and the domain’s convention for a meaningful effect on the auxiliary panel. The threshold cannot be derived because this tradeoff cannot.

Why the four-cell structure survives threshold ambiguity. Different analysts may set τ_{NC} or τ_{PC} more or less conservatively, so a borderline case could land in different cells depending on whose thresholds are used. What does not change is the action each cell prescribes (Table 1): the *kind* of failure fixes the *kind* of response. A flagged NC failure prescribes recalibration and a flagged PC failure prescribes estimator redesign, whether the threshold was drawn strictly or loosely—because the adjudication identifies a qualitative diagnostic state, not a continuous credibility score. Threshold ambiguity is therefore real but bounded: it affects which cell a borderline analysis lands in, not the action that cell prescribes.

Severity within each cell. The verdict is constant within each cell, but the magnitude and character of the prescribed action can vary based on the severity of the detected failure. Table 2 summarises four operational tiers per cell, from minimal failure within sampling noise to structural identification collapse, mapping each to established sensitivity-analysis and calibration machinery [3, 6, 7, 15, 18–20, 22]. The REDESIGN tiers also capture a qualitative failure pattern we term *cascade-inversion failure*: a literature-anchored proximal–distal cascade recovered with its rank ordering reversed. This is distinct from mere

severity — it is not simply a larger PC deviation, but a sign reversal in the recovered ordering — and to our knowledge has not been named as a category in the causal-inference or mediation literatures. It is narrower than inconsistent (suppression) mediation [12], which concerns opposite-signed direct and indirect effects rather than a reversed ordering across a multi-link cascade. That said, PCs that return null or wrong-signed effects are themselves clear grounds for redesign — the cascade-inversion pattern is the graded case between clean recovery and outright PC failure. The REDESIGN severity vocabulary is markedly thinner beyond Tier 2: NC-based confounding correction has a mature calibration toolchain, whereas PC-recovery failure has mostly been treated as a binary validation gate rather than a graded remediation problem.

Table 2. Severity tiers within the RECALIBRATE and REDESIGN cells. The verdict is constant within each cell; severity governs the scale of the prescribed action (*italic*).

Tier	NC failure (Recalibrate)	PC failure (Redesign)
1	Within sampling noise. <i>Report Robustness values and E-value as bounds.</i>	Agreement metrics preserved. <i>Document; release with caveat.</i>
2	Non-null but bounded. <i>Schuemie empirical-null calibration; report $(\hat{\mu}, \hat{\sigma})$.</i>	One or two metrics lost. <i>Estimator-swap across designs.</i>
3	NC/target ratio 0.3–1.0. <i>Tchetgen control-outcome calibration or proximal CI.</i>	Cascade-inversion. <i>Return to design; check nuisance-model misspecification.</i>
4	NC magnitude approaches target. <i>Sharp partial-ID bounds; re-specify identification.</i>	PCs null or wrong-signed. <i>Abandon, adopt IV, or re-scope.</i>

Two analytical approaches, one decision rule. JANUS sits between two analytical traditions that must not be conflated. The OHDSI tradition is panel-based and parametric, asking “what does the residual systematic-error distribution look like for this estimator on this data?” [4, 15]. The Cinelli–Hazlett tradition is single-estimate and model-based, asking “how strong would an unmeasured confounder need to be to overturn this result?” [3]. Both are legitimate and non-interchangeable, and JANUS invokes each at the appropriate severity tier: Cinelli–Hazlett at Tiers 1–2 when a single-estimate sensitivity report is the right documentation; Schuemie empirical-null at Tier 2 when a panel of NCs is available; proximal causal inference at Tier 3 when a double-NC structure supports bridge-function identification [14, 19]; re-specification at Tier 4 when neither tradition’s tools apply. The contribution is the structured mapping from joint NC/PC signature to the appropriate tool, not the underlying machinery.

Technical constraints. Three constraints on the severity machinery deserve direct acknowledgement. The Expected Absolute Systematic Error (EASE) < 0.25 threshold at Tier 2 is presented by Conover et al. as a proposed threshold rather than a derived bound [4]. The validity of NCs as confounding detectors is contested: Penning-de Vries et al. argue they may lack both specificity and sensitivity [21], and Zafari et al. find that across 184 pharmacoepidemiologic studies NCs were used primarily for bias detection rather than correction [25].

Synthesised PCs are epistemically different from the literature-anchored PCs JANUS uses in Section 4: the former test whether the estimator scales with a known injected magnitude, the latter whether it recovers a meta-analytic anchor. None of these constraints invalidates the framework, but all sharpen its operating envelope.

4 Empirical Demonstration: SSB and Type 2 Diabetes

We illustrate the JANUS decision on a sugar-sweetened-beverage (SSB) and type-2-diabetes analysis using double machine learning at Middle-layer Super Output Area (MSOA) level across England—a UK census geography of roughly 5,000–15,000 residents each—developed in an upcoming companion paper. The auxiliary panel spans the wider metabolic cascade, but the target estimand is the average causal effect of SSB consumption on type 2 diabetes (T2DM) prevalence alone. Retail grocery purchasing data provide exposure measurement; Quality and Outcomes Framework (QOF) prevalence at MSOA level provides outcome measurement—QOF is a national programme under which general practices report standardised disease-register counts, yielding area-level prevalence for conditions such as T2DM. The primary panel comprises six biologically plausible outcomes: the T2DM target; obesity and hypertension, literature-established positive controls grounded in the metabolic cascade from adiposity to diabetes [13, 24]; and three metabolically adjacent outcomes expected to be attenuated relative to these — non-diabetic hyperglycaemia, coronary heart disease, and chronic kidney disease — which enter the primary-magnitude summary but are not used as anchored positive controls, as clean external effect estimates are unavailable for them. Five negative controls — learning disabilities, epilepsy, rheumatoid arthritis, osteoporosis, and dementia — are chosen on subject-matter grounds to lack a plausible causal pathway from SSB consumption while plausibly sharing area-level confounding structure (deprivation gradient, baseline health-behaviour clustering, healthcare-access variation). The analysis at the primary inclusion threshold ($n \geq 50$ sugar-purchasing customers per MSOA, $N = 2,606$) provides the worked example below (Fig. 1); Section 5 exhibits the remaining three JANUS cells through constructed counterfactual variants.

4.1 Target and positive-control panel

The DML target estimate is $\hat{\theta} = 2.69$ percentage points (pp) per g/100 ml SSB consumption on T2DM prevalence. The two literature-anchored positive controls recover effects in the predicted direction and at magnitudes consistent with the metabolic cascade [13, 24]: obesity at $\hat{\beta}_{PC} = 5.30$ pp and hypertension at $\hat{\beta}_{PC} = 3.04$ pp. Both positive controls recover the expected direction and the pre-specified proximal–distal ordering (obesity $\downarrow$ hypertension $\downarrow$ diabetes); under the panel-level recovery criterion, the PC panel therefore passes. The three further biologically plausible outcomes return positive effects of decreasing magnitude

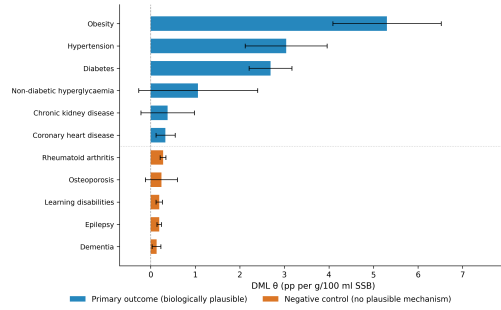


Fig. 1. DML estimates for the negative-control and primary-outcome panels at the primary inclusion threshold ($n \geq 50$, $N = 2,606$). Orange: five negative controls (no plausible causal pathway from SSB); blue: six biologically plausible primary outcomes, including the diabetes target and the obesity/hypertension positive controls. Error bars are 95% CIs. NC estimates are uniformly positive and small (0.13–0.28 pp); the primary panel recovers the expected proximal–distal ordering (obesity > hypertension > diabetes). Joint signature: NC fail, PC pass $\Rightarrow$ RECALIBRATE.

with proximity to the metabolic core: non-diabetic hyperglycaemia at 1.06 pp, chronic kidney disease at 0.38 pp, and coronary heart disease at 0.33 pp. Across all six primary outcomes the estimator reproduces the expected proximal–distal ordering implied by the structure of metabolic-disease causation, with obesity and hypertension the strongest and the more distal conditions attenuated. The PC panel passes.

4.2 Negative-control panel and the joint signature

The negative-control panel returns a coherent positive signature across all five controls: learning disabilities at $\hat{\beta}_{NC} = 0.19$ pp, epilepsy at 0.19 pp, rheumatoid arthritis at 0.28 pp, osteoporosis at 0.24 pp, and dementia at 0.13 pp. The NC estimates are individually small but uniformly positive and of similar magnitude. Under random sampling variation around a true null, the five estimates would scatter around zero with mixed signs; their coherent positive pattern instead indicates a systematic upward bias shared across the panel, consistent with residual confounding rather than chance.

Two diagnostics characterise this signature. First, the mean primary-outcome magnitude exceeds the mean negative-control magnitude by a factor of $10.3\times$ at the primary threshold, and this separation is stable across inclusion thresholds ($10.1\times$ at $n \geq 10$ through $11.5\times$ at $n \geq 100$); the ratio is a descriptive effect-size summary rather than an inferential test. Second, a one-sided label-permutation test (10,000 permutations over the pooled eleven-outcome panel) that randomly reassigns the primary/negative-control labels and compares the difference in mean $|\hat{\theta}|$ between the two groups rejects the null of exchangeable magnitudes at $p = 0.0018$, establishing that the primary–NC separation is not attributable to chance. Under the panel-level JANUS criteria (Section 3.1), the coherent NC

deviation yields NC fail, while recovery of the expected positive-control direction and proximal–distal ordering yields PC pass; the verdict is **Recalibrate**.

4.3 Action prescription and corrected estimate

The Recalibrate cell prescribes bias correction informed by the NC signature, characterisation of the bias mechanism through root-cause attribution, and release of the corrected estimate with explicit residual-bias reporting. Applying the basic correction $\hat{\theta}_{\text{corrected}} = \hat{\theta} - \hat{\beta}_{\text{NC}}$ under the assumption that the bias acts identically on target and NCs [11] subtracts the mean NC effect (0.21 pp) from the target, yielding $\hat{\theta}_{\text{corrected}} = 2.48$ pp (2.69×0.923) — a 7.7% reduction. The corrected estimate remains highly significant.

The bias-equivalence assumption underwriting this correction is strong: it requires the unmeasured confounding bundle affecting T2DM to affect the NCs at the same average magnitude. It is partially defensible, since the NCs were chosen for shared confounding structure with the target — area-level deprivation, baseline health-behaviour clustering, and healthcare-access variation plausibly affect all outcomes through related mechanisms — but it is not verifiable from the data alone. The 7.7% correction is therefore reported not as a precision claim about the true effect but as a conservative-robustness operation: under approximate bias-equivalence, the corrected estimate is a lower bound on the SSB→T2DM effect that survives the residual confounding the NC panel has detected. Where larger NC panels are available, more sophisticated calibrations become viable; Section 3.3 maps these to severity tiers within the Recalibrate cell.

4.4 Honest framing of root-cause attribution

The RECALIBRATE action prescription commits the analyst to characterising the bias mechanism through root-cause analysis. In the present analysis, root-cause attribution is candidate-enumeration rather than candidate-verification: area-level deprivation, baseline metabolic-prevalence variation, retail-density confounding, and selection in the specific retail shopping population are all plausible contributors to the NC signature, and the bias correction is conservative across this candidate set; but the data do not identify which mechanism is doing how much of the work.

5 Pedagogical Counterfactuals

Section 4 demonstrated the empirical Recalibrate cell. Figure 2 places this result alongside constructed variants for Release, Redesign, and Withhold using the same eleven-outcome panel. These counterfactuals illustrate the diagnostic signatures associated with each verdict rather than claiming statistical guarantees under arbitrary identification failures.

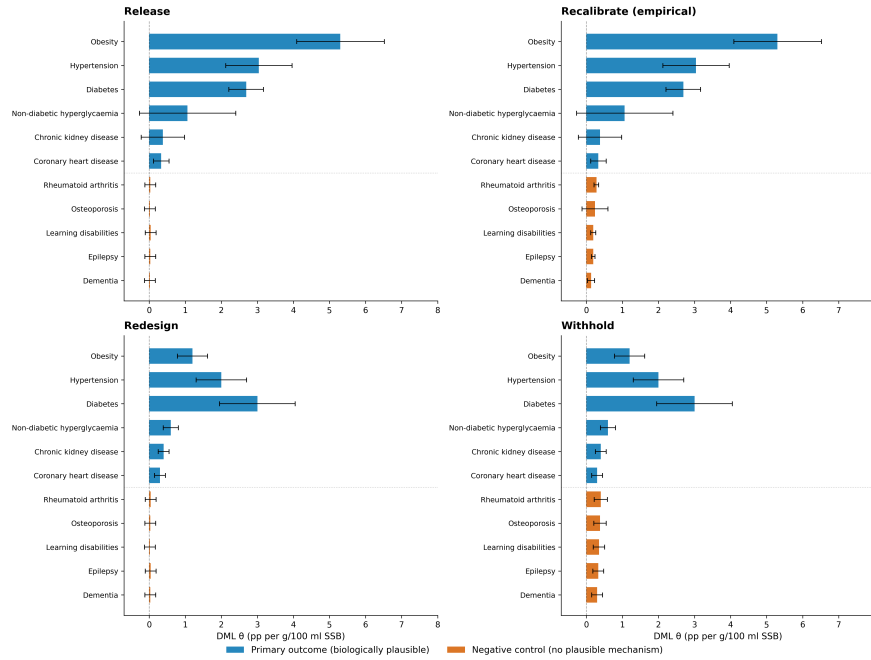


Fig. 2. The four JANUS verdict cells across the same eleven-outcome panel (six primary-panel outcomes and five negative controls). *Top row:* PC-pass (RELEASE, RECALIBRATE); *bottom row:* PC-fail (REDESIGN, WITHHOLD). *Left:* NC-pass; *right:* NC-fail. The RECALIBRATE panel (top-right) is the empirical result (Section 4); the other three are constructed illustrations. Outcomes are held in fixed row order (the correct proximal–distal cascade), so a recovered cascade reads as a clean descending staircase and estimator failure as a broken ordering. The RECALIBRATE/WITHHOLD difference is set by the PC signature, not NC magnitude alone.

Release (NC pass, PC pass). The top-left panel shows the joint signature consistent with correct identification. All five negative controls return effects indistinguishable from zero (each within 0.05 pp), and the primary panel recovers the literature-anchored metabolic cascade in the expected magnitude ordering (obesity > hypertension > diabetes, with the more distal sequelae attenuated). The verdict is RELEASE: the estimator produces no confounding signature on outcomes for which a non-zero effect would be implausible, and recovers the literature-anchored cascade within tolerance on outcomes for which substantial prior evidence exists. The prescribed action is to report the diabetes estimate as decision-grade with standard scientific caveats.

Redesign (NC pass, PC fail). The bottom-left panel shows estimator failure without confounding signature. The negative controls return effects near zero — no evidence of unmeasured-confounder bias on outcomes that share the target’s confounding structure — but the primary panel does not recover the literature-anchored cascade: the ordering is inverted, with diabetes estimated as

the strongest effect and obesity weakest at 1.20 pp, well below its 5.30 pp literature anchor. The inversion is the key diagnostic: a correctly specified estimator should recover the proximal–distal structure of metabolic-disease causation, and one returning the inverse ordering cannot be assumed to recover the unknown target effect either, even with no confounding signature on the NC panel. The verdict is REDESIGN: the failure is on the estimator side, and the prescribed action is return-to-design. Possible causes include nuisance-model misspecification, inadequate covariate adjustment, or design-assumption violations such as SUTVA failure under spatial interference between adjacent MSOAs [17].

Withhold (NC fail, PC fail). The bottom-right panel shows compounded failure. The negative controls return effects in the 0.30–0.40 pp band — elevated above the empirical RECALIBRATE baseline and well above the RELEASE counterfactual’s zero band — indicating a confounding signature on outcomes for which no effect is plausible. Simultaneously, the primary panel shows the same inverted cascade as the REDESIGN counterfactual: obesity at 1.20 pp against a 5.30 pp anchor, hypertension at 2.00 pp against a 3.04 pp anchor. Both checks fail. The verdict is WITHHOLD: $\hat{\theta}$ should not be released as a credible causal effect under the stated assumptions. This concerns the causal claim, not the analysis’s scientific value—a joint control failure is itself informative, and reporting it transparently as exploratory, with the diagnostics shown, remains legitimate. What Withhold precludes is presenting $\hat{\theta}$ as an identified effect.

The required redesign is substantive, and the signature indicates its direction. A uniform NC failure points either to an unblocked confounding path (a revised adjustment set) or to mis-specified negative controls (independent validation); an attenuated PC cascade points to estimator misspecification or to literature anchors that do not transfer at MSOA level (re-examination of anchors and the exposure proxy). Distinguishing these requires the diagnostic work of Section 3.1; the verdict identifies that redesign is needed and constrains its direction without prescribing a single fix.

The contrast with the empirical RECALIBRATE panel is pedagogically important: the empirical NC magnitudes (0.13–0.28 pp) are not categorically different from the WITHHOLD counterfactual’s (0.30–0.40 pp), but the empirical primary panel preserves the literature-anchored cascade while the WITHHOLD panel does not. The RECALIBRATE/WITHHOLD verdict difference is therefore set by the PC signature, not the NC signature alone — a structural feature of the joint adjudication that single-signal rules cannot replicate.

6 Discussion

6.1 Application across regimes

The JANUS decision’s structural form is universal, but the verdicts available and the calibration of $\tau_{\text{NC}}, \tau_{\text{PC}}$ depend on the application domain’s cost structure (Table 3).

Table 3. JANUS adjudication across application domains. Cost-structure asymmetry determines which verdicts are available and how thresholds are set.

Domain	NC	PC	Verdict	Cost caveat
Population health (this paper)	Implausible DV	Cascade outcome	Release / Recal.	Confounding inevitable; recal. if characterised
Pharmaco-epi. [4, 15]	50–100 NCs	Synthesised	Recalibrate	Regulated; calibrated false-flag
Recommender [9, 23]	A/A expts	Vetted expts	Release; off-diag. RCA	Tightened ($p < 0.0005$)
Safety eng. [10]	1002 voter	Fault sigs	Withhold	Catastrophic; recal. forbidden

6.2 The recalibrate verdict as symptom and cause

The recalibrate verdict is structured around a distinction between a diagnostic signal and its underlying cause. A negative-control failure provides evidence that the target estimate is biased, while the severity tier determines the corresponding correction, calibrated to the magnitude of the confounding the failure implies. In this respect JANUS shares the logic of anomaly attribution found in other applied literatures, in which a detected irregularity is traced to its generating mechanism. The orientation here is, however, inverted. Rather than invoking causal structure to account for an observed anomaly, JANUS treats anomalies in a panel of auxiliary causal estimates as the diagnostic instrument itself, using their joint behaviour to adjudicate the credibility of the target estimate.

6.3 Open directions

The framework adjudicates a single estimate at one point in time; three extensions relax this. Applied sequentially as data accumulate, JANUS could also detect changes in identification status: transitions between verdicts may signal shifts in confounding or estimator performance, while an NC failure preceding PC failure could provide an early indicator of broader identification breakdown.

The fixed tolerance τ_{NC} could be replaced by a conformal prediction region, which sets the acceptance bound adaptively with a coverage guarantee [1], at the cost of a larger calibration panel. A formal treatment of how to set the thresholds—balancing the false-flag and false-pass errors of Section 3.3 against the costs of the downstream decision—is the natural next step.

7 Conclusion

The JANUS decision is a synthesis, not a mechanism. Negative controls, positive controls, empirical confidence-interval calibration, redundant sensor cross-checking, and A/B pre-testing are each mature in their home disciplines, but no tradition has unified their joint pass/fail logic into a single pre-specified decision rule returning a structured verdict over an individual observational causal estimate. JANUS supplies that rule: the four-verdict structure encodes a qualitative

map of identification status robust to the threshold ambiguity affecting boundary cases, since the kind of failure determines the prescribed action regardless of where tolerances are drawn, while the severity tiers map failure magnitude to established sensitivity-analysis and calibration machinery. Its intended audience is researchers entering observational causal inference from applied backgrounds, for whom the OHDSI empirical-calibration pipeline is not standard training but the need to adjudicate estimates against auxiliary signals is increasingly pressing. The rule is teachable precisely because the action prescription follows from which quadrant of the two-by-two the joint NC/PC signature occupies — a structure any analyst can read off a single figure.

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